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(1) $f(z) = z^2$ identificar u y v / Cauchy-Riemann es diferenciable en todas partes

Sol:

$z = x + iy$

$f(z) = (x+iy)^2 = x^2 + 2ixy + i^2y^2 = (x^2-y^2) + 2ixy$

$\Rightarrow \begin{cases} u(x,y) = x^2 - y^2 \\ v(x,y) = 2xy \end{cases}$ / C-R (1) $\dfrac{\partial u}{\partial x} = \dfrac{\partial v}{\partial y}$

$\Rightarrow \dfrac{\partial u}{\partial x} = 2x \; ; \; \dfrac{\partial v}{\partial y} = 2x \Rightarrow 2x = 2x$

C-R (2) $\dfrac{\partial u}{\partial y} = -\dfrac{\partial v}{\partial x}$

$\dfrac{\partial u}{\partial y} = -2y \; ; \; -\dfrac{\partial v}{\partial x} = -2y \Rightarrow -2y = -2y$

$\therefore$ Rpta: A) Sí es diferenciable en todas partes

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(2) $f(z) = |z|^2$

Sol: $z = x + iy$

$f(z) = |z|^2 = |x+iy|^2 = x^2 + y^2$

$\Rightarrow \begin{cases} u(x,y) = x^2+y^2 \\ v(x,y) = 0 \end{cases}$ / C-R (1) $\dfrac{\partial u}{\partial x} = \dfrac{\partial v}{\partial y}$

$\dfrac{\partial u}{\partial x} = 2x \; ; \; \dfrac{\partial v}{\partial y} = 0 \Rightarrow 2x = 0 \Rightarrow$ Se cumple si $x = 0$

C-R (2) $\dfrac{\partial u}{\partial y} = -\dfrac{\partial v}{\partial x}$

$\dfrac{\partial u}{\partial y} = 2y \; ; \; -\dfrac{\partial v}{\partial x} = 0 \Rightarrow 2y = 0 \Rightarrow$ Se cumple si $y = 0$

$\therefore$ Rpta B) No satisface salvo si $x = y = 0$

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③ $f(z) = \frac{1}{z}$

Sol:

$z = re^{i\theta} \Rightarrow f(z) = \frac{1}{z} = \frac{1}{r}e^{-i\theta} = \frac{1}{r}(\cos\theta - i\,\text{sen}\,\theta)$

$\Rightarrow \begin{cases} u(r,\theta) = \frac{\cos\theta}{r} \\ v(r,\theta) = -\frac{\text{sen}\,\theta}{r} \end{cases}$ | (1) Polar: $\frac{\partial u}{\partial r} = \frac{1}{r}\cdot\frac{\partial v}{\partial \theta}$

$\Rightarrow \frac{\partial u}{\partial r} = -\frac{\cos\theta}{r^2}$; $\frac{1}{r}\frac{\partial v}{\partial\theta} = \frac{1}{r}\cdot\left(-\frac{\cos\theta}{r}\right) = -\frac{\cos\theta}{r^2}$

$\Rightarrow -\frac{\cos\theta}{r^2} = -\frac{\cos\theta}{r^2}$ ✓

(2) Polar: $\frac{1}{r}\frac{\partial u}{\partial\theta} = -\frac{\partial v}{\partial r} \Rightarrow \frac{1}{r}\frac{\partial u}{\partial\theta} = \frac{1}{r}\cdot\left(-\frac{\text{sen}\,\theta}{r}\right) = -\frac{\text{sen}\,\theta}{r^2}$

$-\frac{\partial v}{\partial r} = -\left(\frac{\text{sen}\,\theta}{r^2}\right) = -\frac{\text{sen}\,\theta}{r^2} \Rightarrow -\frac{\text{sen}\,\theta}{r^2} = -\frac{\text{sen}\,\theta}{r^2}$ ✓

∴ Rpta: A) Sí cumple las cond. 2do Teorema

④ $f(z) = \frac{1}{z}$; $f'(z)\ \exists?$

Sol:

$z = x + iy$

$f(z) = \frac{1}{z} = \frac{1}{x+iy} = \frac{1}{x+iy}\cdot\frac{x-iy}{x-iy} = \frac{x-iy}{x^2+y^2} = \frac{x}{x^2+y^2} - i\frac{y}{x^2+y^2}$

$\begin{cases} u(x,y) = \frac{x}{x^2+y^2} \\ v(x,y) = \frac{-y}{x^2+y^2} \end{cases}$ C-R(1) $\frac{\partial u}{\partial x} = \frac{\partial v}{\partial y}$

$\frac{\partial u}{\partial x} = \frac{(x^2+y^2) - x\cdot 2x}{(x^2+y^2)^2} = \frac{y^2-x^2}{(x^2+y^2)^2}$

$\frac{\partial v}{\partial x} = -\frac{(x^2+y^2) - y\cdot 2y}{(x^2+y^2)^2} = -\frac{x^2-y^2}{(x^2+y^2)^2} = \frac{y^2-x^2}{(x^2+y^2)^2}$

$\frac{y^2-x^2}{(x^2+y^2)^2} = \frac{y^2-x^2}{(x^2+y^2)^2}$ ✓

CR(2) $\frac{\partial u}{\partial y} = -\frac{\partial v}{\partial x}$; $\frac{\partial u}{\partial y} = \frac{0 - x\cdot 2y}{(x^2+y^2)^2} = -\frac{2xy}{(x^2+y^2)^2}$

$-\frac{\partial v}{\partial x} = -\frac{0 - y\cdot 2x}{(x^2+y^2)^2} = -\frac{2xy}{(x^2+y^2)^2}$

$\frac{-2xy}{(x^2+y^2)^2} = \frac{-2xy}{(x^2+y^2)^2}$

$f'(z) = \frac{\partial u}{\partial x} + i\frac{\partial v}{\partial x} = \frac{y^2-x^2}{(x^2+y^2)^2} + i\frac{2xy}{(x^2+y^2)^2} = \frac{y^2-x^2+i2xy}{(x^2+y^2)^2} = \frac{-(x-iy)^2}{(x^2+y^2)^2}$

$\boxed{f'(z) = \frac{-\bar{z}^2}{(z\cdot\bar{z})^2} = \frac{-\bar{z}^2}{z^2\cdot\bar{z}^2} = -\frac{1}{z^2}}$ ∴ Rpta: B) Sí existe f' y su valor es $-\frac{1}{z^2}$

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(5) $f(z) = \dfrac{z-1}{2z+1}$

sol: $f(z) = \dfrac{h(z)}{g(z)}$; $h(z) = z-1 \Rightarrow h'(z) = 1$

$g(z) = 2z+1 \Rightarrow g'(z) = 2$

$$f'(z) = \frac{g(z)\cdot h'(z) - h(z)\cdot g'(z)}{|g(z)|^2} = \frac{(2z+1)(1)-(z-1)(2)}{(2z+1)^2}$$

$$f'(z) = \frac{\cancel{2z}+1-\cancel{2z}+2}{(2z+1)^2} = \frac{3}{(2z+1)^2}$$

$\therefore$ Rpta: c) $\boxed{\dfrac{3}{(2z+1)^2}}$